TEXORA · LECTURE SHEET

Properties and Characterization of Polymers

How polymer samples are described and measured: the four molecular-weight averages, the methods that find them, and the thermal, mechanical and structural properties that decide whether a polymer can become a fibre.

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1. Molecular weight averages

A polymer sample contains chains of many lengths, so "the molecular weight" must be an average — and different measuring methods weight the chains differently, giving different averages that describe the same sample.

1.1 Number-average molecular weight, M_n

Count every chain equally: total mass divided by total number of chains.

$$M_n = \Sigma\, N_i M_i \div \Sigma\, N_i$$

Measured by methods that count molecules: **end-group analysis** (titrating the –COOH or $–NH_2$ ends of nylon or polyester) and **osmometry** (osmotic pressure of a solution). Sensitive to the short chains, because they contribute many molecules.

1.2 Weight-average molecular weight, M_w

Weight each chain by its mass, so long chains count more:

$$M_w = \Sigma\, N_i M_i^2 \div \Sigma\, N_i M_i$$

Measured by **light scattering**. M_w governs bulk strength properties better than M_n, because the long chains carry most of the load.

1.3 Viscosity-average and z-average

$\mathbf{M_v}$ comes from solution viscosity (§2) and lies between M_n and M_w, close to M_w. $\mathbf{M_z}$ (from ultracentrifugation) weights the longest chains still more heavily: always $M_n \le M_v \le M_w \le M_z$.

1.4 Polydispersity index (PDI)

$$PDI = M_w \div M_n$$

PDI = 1 only for a perfectly uniform sample. Typical step-growth polymers approach PDI ≈ 2 at high conversion; chain-growth polymers range from about 1.5 (controlled processes) to 5 or more. A broad distribution means troublesome short chains (weak, extractable) and over-long chains (hard to melt and spin) in the same pot.

WORKED EXAMPLE — COMPUTING M_N, M_W AND PDI

A sample contains 2 chains of M = 10,000, 3 chains of M = 20,000 and 1 chain of M = 40,000 g/mol.

$M_n = (2\times10{,}000 + 3\times20{,}000 + 1\times40{,}000) \div (2+3+1) = 120{,}000 \div 6 =$ **20,000 g/mol**.

$M_w = (2\times10{,}000^2 + 3\times20{,}000^2 + 1\times40{,}000^2) \div 120{,}000 = (2\times10^8 + 1.2\times10^9 + 1.6\times10^9) \div 120{,}000 = 3.0\times10^9 \div 120{,}000 =$ **25,000 g/mol**.

PDI = 25,000 ÷ 20,000 = **1.25**. Note how the single 40,000 chain — one chain in six — drags M_w a full 25 % above M_n: weight-averaging always listens to the heavy chains.

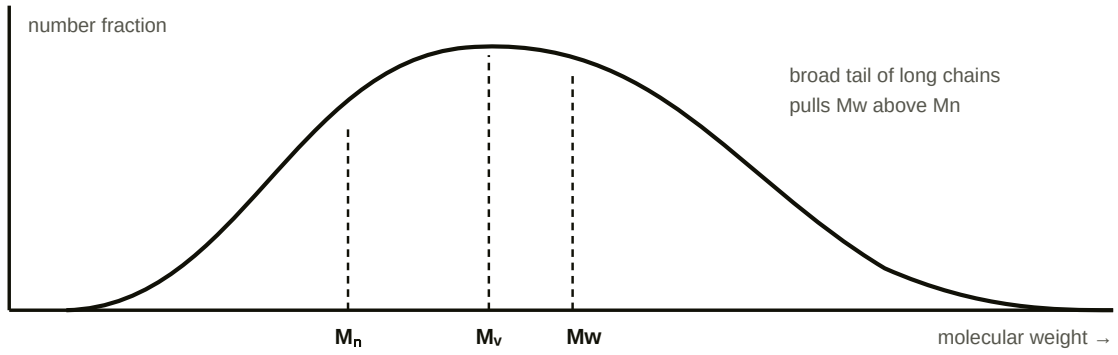


Figure 1 — A molecular-weight distribution curve and the positions of the averages: Mn sits nearest the peak by number, Mw is displaced toward the heavy tail. The wider the curve, the larger the PDI.

2. Measuring molecular weight in practice

METHOD	AVERAGE FOUND	PRINCIPLE	REMARKS
End-group analysis	M_n	Titrate chain-end groups (–COOH with alkali, $–NH_2$ with acid)	Simple, standard for nylon/PET control; fails at very high M where ends are too few to titrate

Membrane osmometry	M_n	Osmotic pressure of solution across a semi-permeable membrane	Absolute method, mid-range M
Solution viscometry	M_v	Flow time of dilute solution vs pure solvent gives [η]; Mark–Houwink converts to M	Cheapest routine method — the PET industry's intrinsic viscosity (IV) is exactly this
Light scattering	M_w	Intensity of light scattered by the solution	Absolute; needs very clean solutions
Gel permeation chromatography (GPC/SEC)	Whole distribution	Solution passes through porous gel; big chains elute first, small chains are delayed in the pores	Gives Mn, Mw and PDI in one run against calibration standards

Intrinsic viscosity [η] is the limiting viscosity number of a solution at zero concentration, and the **Mark–Houwink equation** links it to molecular weight for a given polymer–solvent–temperature system:

$$[\eta] = K \cdot M_v^{\,a}$$

WORKED EXAMPLE — MARK–HOUWINK

For a polyester sample, [η] = 0.75 dL/g measured in a solvent system with $K = 2.1 \times 10^{-4}$ dL/g and $a = 0.82$.

$M_v = ([\eta] \div K)^{1/a} = (0.75 \div 2.1 \times 10^{-4})^{1 \div 0.82} = (3571)^{1.2195} \approx$ **21,500 g/mol**.

Check: $K \cdot M^a = 2.1 \times 10^{-4} \times 21{,}500^{0.82} = 0.75$ dL/g ✓. This is why a spinning plant can control polymer quality with a viscometer: one flow-time measurement stands in for the whole molecular weight.

3. Thermal properties

- **Glass transition temperature, T_g** — the temperature at which the amorphous regions change from a hard glass to a rubbery state as chain segments gain mobility. Below T_g the fibre is stiff and brittle; above it, dye molecules can enter (dyeing of PET needs temperatures above its T_g of about 70–80 °C, or a carrier/plasticiser). Typical values: PET ≈ 70–80 °C, nylon 6,6 ≈ 50 °C (dry; water plasticises it lower), acrylic ≈ 90–100 °C.
- **Melting temperature, T_m** — where the crystallites melt; a range rather than a point, because crystallites differ in size and perfection. Sets the ceiling for ironing, heat-setting and melt spinning (extrusion runs about 20–40 °C above T_m).

- **Heat setting and thermal shrinkage** — heating a drawn fibre near T_m lets stressed chains relax and new crystallites lock the dimensions; an unset fabric shrinks in hot water or ironing instead.
- **Decomposition** — cellulose and PAN degrade before they melt, which is why neither can be melt-spun; thermal stability in the melt (minutes at 280–290 °C) is itself a PET quality criterion.

Characterisation methods: DSC (differential scanning calorimetry) records heat flow versus temperature and shows T_g, crystallisation and T_m in one scan, plus percentage crystallinity from the melting enthalpy; **TGA** (thermogravimetric analysis) tracks weight loss on heating, revealing moisture, volatiles and decomposition onset; **DMA** follows stiffness and damping with temperature.

4. Structural characterisation

- **Crystallinity** — measured by X-ray diffraction (sharp crystal reflections over an amorphous halo), by density (crystals pack tighter: a density-gradient column gives the crystalline fraction), by DSC, and qualitatively by birefringence.
- **Orientation** — the degree to which chains align with the fibre axis, measured optically as **birefringence** (Δn, difference of refractive indices along and across the fibre) or by X-ray azimuthal scans. Orientation, more than crystallinity, is what drawing adds.
- **Chemical structure** — infrared spectroscopy (FTIR) identifies functional groups and polymer type (the standard tool for "which fibre is this?"); NMR resolves detailed structure and comonomer content.

5. Other key properties

PROPERTY	WHAT IT MEANS	TEXTILE CONSEQUENCE
Density	Mass per unit volume (g/cm³): PP 0.91, nylon 1.14, acrylic ≈ 1.17, PET 1.38, viscose ≈ 1.50	Light fibres give more coverage per kilogram; PP floats, PET sinks
Moisture regain	Water absorbed from standard atmosphere, % of oven-dry mass; set by polar groups and amorphous fraction	Comfort, static, dye uptake, dimensional stability
Mechanical	Strength, modulus, elongation, toughness, creep, abrasion	Decide end use: high modulus for cord, high abrasion resistance for carpets
Chemical resistance	Attack by acids, alkalis, solvents, oxidants	Processing windows: PET fears hot strong alkali; nylon fears strong acid
Solubility	Which liquids dissolve the polymer	Chooses the spinning solvent and cleaning methods; basis of fibre

identification tests

WORKED EXAMPLE — MOISTURE REGAIN

A nylon sample weighs 100.0 g after conditioning at 20 °C and 65 % RH, and 96.5 g after oven drying.

Regain = (100.0 − 96.5) ÷ 96.5 × 100 = **3.6 %** — consistent with nylon's tabulated regain of about 4 %, and confirming the sample is not, say, polyester (0.4 % would mean a conditioned weight of only about 96.9 g on the same dry mass). A single weighing thus acts as a rough fibre-identification test.

Glossary

Polydispersity index (PDI) — M_w/M_n, the breadth of the molecular-weight distribution. **Intrinsic viscosity [η]** — the extrapolated viscosity number of a solution at zero concentration. **T_g** — glass transition temperature of the amorphous phase. **T_m** — melting range of the crystallites. **Birefringence** — the difference in refractive index along and across a fibre; a measure of orientation. **DSC / TGA** — calorimetric and gravimetric thermal analysis methods. **GPC** — gel permeation chromatography, separating chains by size in solution.

Key points

• Mn counts chains, Mw weighs them; PDI = Mw/Mn measures the spread — Mn ≤ Mv ≤ Mw ≤ Mz.

• Routine molecular-weight control in fibre plants is by solution viscosity (IV) through the Mark–Houwink relation.

• Tg controls dyeability and stiffness; Tm sets heat-setting, ironing and extrusion limits; PAN and cellulose decompose before melting.

• Crystallinity (X-ray, density, DSC) and orientation (birefringence) together describe a fibre's internal structure.

• Density and regain are quick, decisive characterisation numbers — regain alone often identifies a fibre family.

Practice

1. A sample has 5 chains of 8,000, 5 chains of 12,000 and 2 chains of 30,000 g/mol. Find Mn, Mw and PDI.
2. Why does end-group analysis become unreliable for very high molecular weights?

3. A PET chip has [η] = 0.64 dL/g with K = 2.1 × 10^{-4} and a = 0.82. Estimate Mv. (Take 0.64 ÷ 2.1×10^{-4} ≈ 3048; $3048^{1.2195}$ ≈ 17,800.)
4. Explain why PET fabric is dyed at 130 °C under pressure, or at the boil with a carrier, but never satisfactorily at 80 °C in plain water — answer using Tg.
5. Two nylon samples have the same Mn but PDIs of 1.8 and 3.5. Which will spin more uniformly, and why?

Answer key

1. Total mass = 5×8,000 + 5×12,000 + 2×30,000 = 40,000 + 60,000 + 60,000 = 160,000; chains = 12. Mn = 160,000 ÷ 12 ≈ **13,333**. Mw = (5×$8{,}000^2$ + 5×$12{,}000^2$ + 2×$30{,}000^2$) ÷ 160,000 = (3.2×10^8 + 7.2×10^8 + 1.8×10^9) ÷ 160,000 = 2.84×10^9 ÷ 160,000 = **17,750**. PDI = 17,750 ÷ 13,333 ≈ **1.33**.
2. The number of end groups per gram falls as 1/M; at high M the titre becomes so small that titration error swamps the signal — the method "counts" ends that are barely there.
3. Mv ≈ **17,800 g/mol** — a low-side chip, suitable for textile filament rather than high-tenacity or bottle grades.
4. Below Tg (≈ 70–80 °C) the amorphous regions of PET are glassy and their chain segments cannot move aside to admit dye molecules; at 80 °C the fibre is only just at the transition and water does not plasticise PET. At 130 °C the segments are fully mobile and disperse dye diffuses in; a carrier plasticises the surface to the same effect at the boil.
5. The PDI 1.8 sample. A narrow distribution melts and flows uniformly, draws evenly and has few short chains acting as weak points; the PDI 3.5 sample contains many short chains (weakness, stickiness) and over-long chains (high local viscosity, filament breaks and uneven draw).
